

Statistical Inference Report

Fintech Loan Application A/B Test

Inferential Statistics and Hypothesis Testing

1 Objective

A fintech company is testing a new streamlined loan application process against its current process. The objective is to determine whether the new process improves efficiency without negatively affecting approval outcomes.

2 Dataset Description

The dataset contains 200 applicants randomly assigned to one of two groups:

- `Current_Process`
- `Streamlined_Process`

Table 1: Variable Definitions

Variable	Description
<code>applicant_id</code>	Unique applicant identifier
<code>group</code>	Application process used
<code>time_to_complete_min</code>	Completion time (minutes)
<code>approved</code>	Loan approval status (Yes/No)
<code>credit_score</code>	Applicant credit score
<code>loan_amount_requested</code>	Requested loan amount (USD)

3 Statistical Tasks

1. Independent Samples t-test

Compare the mean completion time between the two groups.

Report:

- Mean completion time for each group
- t-statistic
- Degrees of freedom
- p-value

2. Confidence Interval

Construct a 95% confidence interval for the difference in mean completion time.

Interpret the result in plain language.

3. Effect Size

Calculate Cohen's d for the difference in completion time.

Classify the effect size as:

- Small
- Medium
- Large

Discuss whether the difference is practically significant.

4. Chi-Square Test of Independence

Determine whether approval rates differ between the two process groups.

Report:

- χ^2 statistic
- Degrees of freedom
- p-value
- Cramer's V

5. Assumption Checking

Verify:

- Normality using Shapiro-Wilk test or Q-Q plots
- Homogeneity of variances using Levene's test

If assumptions are violated, justify and apply an appropriate alternative test.

4 Discussion

Answer the following questions:

1. Is the streamlined process significantly faster?
2. Does the streamlined process affect approval rates?
3. Is the observed difference practically meaningful?
4. Based on the statistical evidence, would you recommend adopting the streamlined process?
Justify your answer.

5 Deliverables

1. A fully commented analysis script (Python).
2. Tables and visualizations supporting each statistical test.
3. A concise report (2–3 pages) summarizing methods, results, and recommendations.